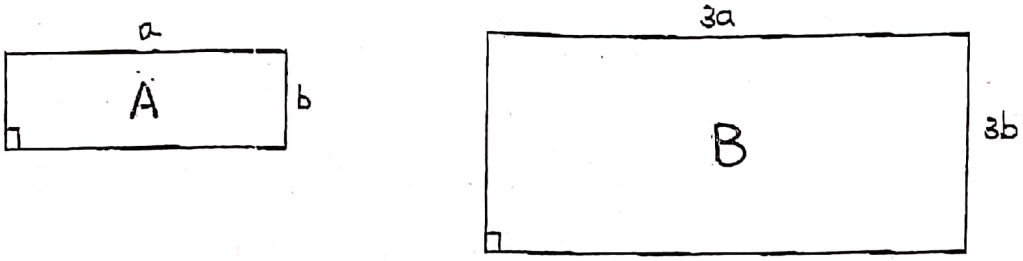
Scale factor investigation Memo

Similar shapes

The following shapes are similar

1)

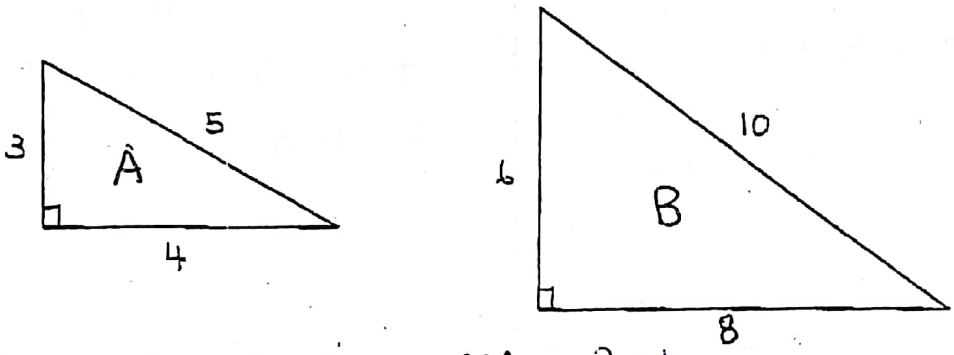


a) Write down the length scale factor $B:A = 3:1$

b) Write down the area scale factor:

$$\frac{\text{Area } B}{\text{Area } A} = \frac{3a \times 3b}{a \times b} = 9$$

2)

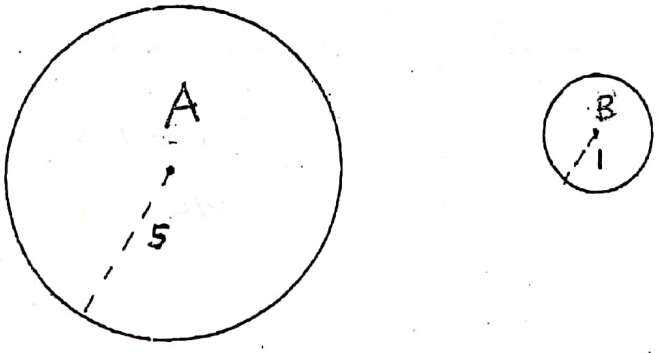


a) Write down the length scale factor $B:A = 2:1$

b) Write down the area scale factor:

$$\frac{\text{Area } B}{\text{Area } A} = \frac{\frac{1}{2} 8 \times 6}{\frac{1}{2} 4 \times 3} = 4$$

3)



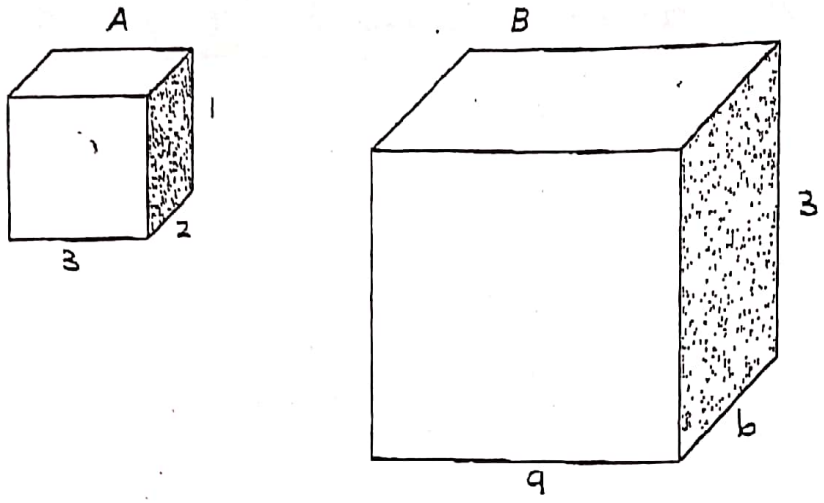
a) Write down the length scale factor $A:B: = 5:1$

b) Write down the area scale factor:

$$\frac{\text{Area } A}{\text{Area } B} = \frac{\pi (5)^2}{\pi (1)^2} = 25$$

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4)



a) Write down the length scale factor B : A = 9 : 3 = 3 : 1

b)

Find the surface area of the small box	Find the surface area of the large box
$SA = 2(3\times1) + 2(2\times1) + 2(3\times2)$ $= 6 + 4 + 12$ $= 22$	$SA = 2(9\times3) + 2(6\times3) + 2(6\times9)$ $= 54 + 36 + 108$ $= 198$

c) Find the area scale factor:

$$\frac{\text{Area of large box (B)}}{\text{Area of small box (A)}} = \frac{198}{22} = 9$$

d)

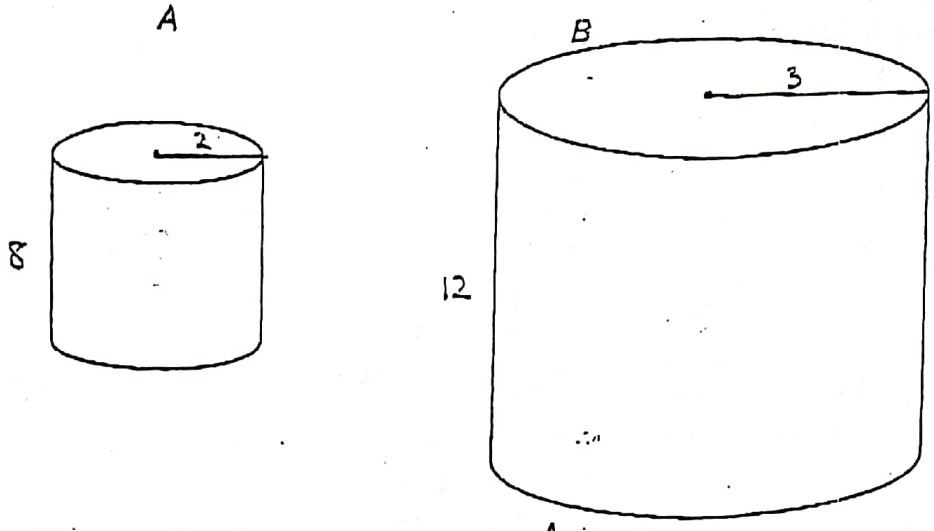
Find the volume of the small box(A)	Find the volume of the large box (B)
$V = 3\times2\times1$ $= 6$	$V = 9\times6\times3$ $= 162$

e) Therefore, the volume scale factor :

$$\frac{\text{Volume of large box (B)}}{\text{Volume of small box (A)}} = \frac{162}{6} = 27$$

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5)



a) Write down the length scale factor $B:A = \frac{3}{2}$

b) Find the surface area of the small cylinder

$2\pi r^2 + 2\pi rh$

$= 2\pi(2)^2 + 2\pi(2)(8)$

$= 40\pi$

Find the surface area of the large cylinder

$2\pi R^2 + 2\pi RH$

$= 2\pi(3)^2 + 2\pi(3)(12)$

$= 90\pi$

c) Find the area scale factor:

$$\frac{\text{Area of large cylinder (B)}}{\text{Area of small cylinder (A)}} = \frac{90\pi}{40\pi} = \frac{9}{4}$$

) Find the volume of the small cylinder(A)

$V = \pi r^2 H$

$= \pi(2)^2(8)$

$= 32\pi$

Find the volume of the large cylinder(B)

$V = \pi R^2 H$

$= \pi(3)^2(12)$

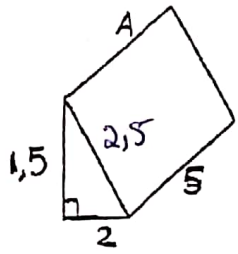
$= 108\pi$

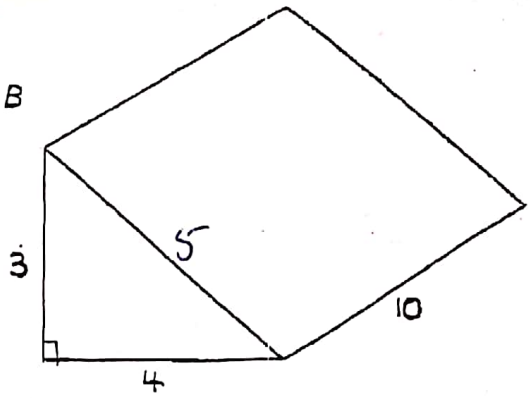
e) Therefore, the volume scale factor :

$$\frac{\text{Volume of large cylinder (B)}}{\text{Volume of small cylinder (A)}} = \frac{108\pi}{32\pi} = \frac{27}{8}$$

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6)





a) Write down the length scale factor B : A 2

b) Find the surface area of each triangular prism

Small

$= 2 \times \left(\frac{1,5 \times 2}{2}\right) + (2,5 \times 5) + (2 \times 5) + (1,5 \times 5)$

$= 33.$

Big

$2 \times \left(\frac{3 \times 4}{2}\right) + 50 + 30 + 40$

$= 132$

c) $\frac{\text{Area of large triangular prism (B)}}{\text{Area of small triangular prism (A)}} = \frac{132}{33} = 4$

d) Find the volume of both triangular prisms

Small

$\text{Vol} = \frac{1}{2} \times b \times h \times lt$

$= \frac{1}{2} \times 2 \times 1,5 \times 5$

$= 7,5$

Big

$\text{Vol} = \frac{1}{2} \times 4 \times 3 \times 10$

$= 60$

e) Therefore, the volume scale factor =

$\frac{\text{Volume of large triangular prism (B)}}{\text{Volume of small triangular prism (A)}} = \frac{60}{7,5} = 8$

Now make a table of all the results:

Shape	1	2	3	4	5	6
Length scale factor	3	2	5	3	3/2	2
Area scale factor	9	4	25	9	9/4	4
Volume scale factor				27	27/8	8

In general:

Length scale factor k

Area scale factor k^2

Volume scale factor k^3

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